In [3]:
```python
import numpy as np
import pandas as pd
```

In [4]:
```python
hospital_data = pd.read_csv('hospital_patient_analytics.csv', index_col='Patient_ID')
```

In [5]:
```python
hospital_data
```

Out[5]:

Patient_ID	Admission_Date	Discharge_Date	Department	Admission_Type	Diagnosis	Severity_Level	Age	Insurance_Type	Length_of_S
PT10001	17/05/2024	01/06/2024	Internal Medicine	Emergency	Pneumonia	Critical	43	HMO	
PT10002	30/08/2024	12/09/2024	General Surgery	Emergency	Tumor Removal	High	23	Medicaid	
PT10003	13/04/2023	16/04/2023	Oncology	Urgent	Colon Cancer	Moderate	79	Medicare	
PT10004	09/12/2024	15/12/2024	Orthopedics	Elective	Sports Injury	Low	44	Medicare	
PT10005	09/04/2024	15/04/2024	Orthopedics	Emergency	Joint Replacement	High	30	HMO	
...	...	...	...	...	...	...	...	...	
PT10226	12/03/2023	16/03/2023	Pediatrics	Elective	Ear Infection	Moderate	17	PPO	
PT10227	13/07/2024	15/07/2024	Orthopedics	Elective	Spinal Injury	Low	18	Medicare	
PT10228	20/10/2024	26/10/2024	Internal Medicine	Urgent	COPD	Moderate	55	Private Insurance	
PT10229	14/11/2023	21/11/2023	Neurology	Urgent	Stroke	Moderate	68	PPO	
PT10230	08/05/2023	12/05/2023	Cardiology	Elective	Hypertension	Moderate	83	Private Insurance	

230 rows × 17 columns

In [6]:
```
hospital_data.info()
```

```
<class 'pandas.DataFrame'>
Index: 230 entries, PT10001 to PT10230
Data columns (total 17 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Admission_Date              230 non-null    str
 1   Discharge_Date              230 non-null    str
 2   Department                  230 non-null    str
 3   Admission_Type              230 non-null    str
 4   Diagnosis                   230 non-null    str
 5   Severity_Level              230 non-null    str
 6   Age                         230 non-null    int64
 7   Insurance_Type              230 non-null    str
 8   Length_of_Stay_Days         230 non-null    int64
 9   Total_Charges               230 non-null    float64
 10  Insurance_Coverage          224 non-null    float64
 11  Wait_Time_Hours             230 non-null    float64
 12  Number_of_Procedures        230 non-null    int64
 13  Staff_to_Patient_Ratio      220 non-null    float64
 14  Patient_Satisfaction_Score  220 non-null    float64
 15  Complications               230 non-null    int64
 16  Readmitted_30Days           230 non-null    int64
dtypes: float64(5), int64(5), str(7)
memory usage: 32.3+ KB
```

I used the mean of the column with missing data to fill up the column

In [7]:
```
hospital_data['Insurance_Coverage'].mean()
```

Out[7]:
```
np.float64(17276.098973214284)
```

In [8]:
```
hospital_data[['Insurance_Coverage', 'Staff_to_patient_Ratio', 'Patient_Satisfaction_Score']] = hospital_data[['Insurance_Cove
```

In [9]:
```
hospital_data.info() # i checked again to be sure all columns are filled up
```

```
<class 'pandas.DataFrame'>
Index: 230 entries, PT10001 to PT10230
Data columns (total 18 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Admission_Date              230 non-null    str
 1   Discharge_Date              230 non-null    str
 2   Department                  230 non-null    str
 3   Admission_Type              230 non-null    str
 4   Diagnosis                   230 non-null    str
 5   Severity_Level              230 non-null    str
 6   Age                         230 non-null    int64
 7   Insurance_Type              230 non-null    str
 8   Length_of_Stay_Days         230 non-null    int64
 9   Total_Charges               230 non-null    float64
 10  Insurance_Coverage          230 non-null    float64
 11  Wait_Time_Hours             230 non-null    float64
 12  Number_of_Procedures        230 non-null    int64
 13  Staff_to_Patient_Ratio      220 non-null    float64
 14  Patient_Satisfaction_Score  230 non-null    float64
 15  Complications               230 non-null    int64
 16  Readmitted_30Days           230 non-null    int64
 17  Staff_to_patient_Ratio      230 non-null    float64
dtypes: float64(6), int64(5), str(7)
memory usage: 34.1+ KB
```

In [10]:
```
hospital_data['Patient_Out_of_Pocket'] = hospital_data['Total_Charges'] - hospital_data['Insurance_Coverage']
```

In [11]:
```
hospital_data.info()
```

```
<class 'pandas.DataFrame'>
Index: 230 entries, PT10001 to PT10230
Data columns (total 19 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Admission_Date              230 non-null    str
 1   Discharge_Date              230 non-null    str
 2   Department                  230 non-null    str
 3   Admission_Type              230 non-null    str
 4   Diagnosis                   230 non-null    str
 5   Severity_Level              230 non-null    str
 6   Age                         230 non-null    int64
 7   Insurance_Type              230 non-null    str
 8   Length_of_Stay_Days         230 non-null    int64
 9   Total_Charges               230 non-null    float64
 10  Insurance_Coverage          230 non-null    float64
 11  Wait_Time_Hours             230 non-null    float64
 12  Number_of_Procedures        230 non-null    int64
 13  Staff_to_Patient_Ratio      220 non-null    float64
 14  Patient_Satisfaction_Score  230 non-null    float64
 15  Complications               230 non-null    int64
 16  Readmitted_30Days           230 non-null    int64
 17  Staff_to_patient_Ratio      230 non-null    float64
 18  Patient_Out_of_Pocket       230 non-null    float64
dtypes: float64(7), int64(5), str(7)
memory usage: 35.9+ KB
```

In [12]:
```
hospital_data['Patient_Out_of_Pocket']
```

Out[12]:
```
Patient_ID
PT10001    13722.69
PT10002    17579.22
PT10003     3118.26
PT10004     6947.35
PT10005     8636.74
             ...
PT10226     2381.43
PT10227     3449.84
PT10228     3665.76
PT10229     3368.04
PT10230     1510.80
Name: Patient_Out_of_Pocket, Length: 230, dtype: float64
```

In [13]:
```
hospital_data.info()
```

```
<class 'pandas.DataFrame'>
Index: 230 entries, PT10001 to PT10230
Data columns (total 19 columns):
 #   Column                      Non-Null Count  Dtype
---  ------                      --------------  -----
 0   Admission_Date              230 non-null    str
 1   Discharge_Date              230 non-null    str
 2   Department                  230 non-null    str
 3   Admission_Type              230 non-null    str
 4   Diagnosis                   230 non-null    str
 5   Severity_Level              230 non-null    str
 6   Age                         230 non-null    int64
 7   Insurance_Type              230 non-null    str
 8   Length_of_Stay_Days         230 non-null    int64
 9   Total_Charges               230 non-null    float64
 10  Insurance_Coverage          230 non-null    float64
 11  Wait_Time_Hours             230 non-null    float64
 12  Number_of_Procedures        230 non-null    int64
 13  Staff_to_Patient_Ratio      220 non-null    float64
 14  Patient_Satisfaction_Score  230 non-null    float64
 15  Complications               230 non-null    int64
 16  Readmitted_30Days           230 non-null    int64
 17  Staff_to_patient_Ratio      230 non-null    float64
 18  Patient_Out_of_Pocket       230 non-null    float64
dtypes: float64(7), int64(5), str(7)
memory usage: 44.0+ KB
```

I grouped the data by department to see their patient size, length of stay, total revenue and satisfaction

In [14]:
```
hospital_data.groupby('Department').agg(
Patient_Count=('Department', 'size'),
Avg_Length_of_Stay=('Length_of_Stay_Days', 'mean'),
Total_Revenue=('Total_Charges', 'sum'),
Avg_Satisfaction=('Patient_Satisfaction_Score', 'mean'))
```

Out[14]:

Department	Patient_Count	Avg_Length_of_Stay	Total_Revenue	Avg_Satisfaction
Cardiology	39	6.897436	1090912.88	7.410967
Emergency	32	8.468750	1087010.51	6.552599
General Surgery	29	7.413793	991504.25	7.297508
Internal Medicine	27	6.148148	690842.59	7.552879
Neurology	28	5.392857	731747.75	7.408133
Oncology	34	5.617647	847936.33	6.947874
Orthopedics	23	7.130435	607103.59	7.427292
Pediatrics	18	5.111111	333039.41	7.423763

I also grouped the data by Admission type to see their mean charge, length of stay and wait time

In [15]:
```
df = hospital_data.groupby('Admission_Type').agg(
Avg_Charge=('Total_Charges', 'mean'),
Avg_Length_of_Stay=('Length_of_Stay_Days', 'mean'),
Avg_wait_time=('Wait_Time_Hours', 'mean'))
df = df.sort_values(by='Avg_Charge', ascending=False)

df[['Avg_Charge','Avg_Length_of_Stay', 'Avg_wait_time']] = df[['Avg_Charge','Avg_Length_of_Stay', 'Avg_wait_time']].round(2)
print(df)
```

```
                Avg_Charge  Avg_Length_of_Stay  Avg_wait_time
Admission_Type
Emergency          42632.58               9.95           4.68
Urgent             21719.01               5.28           2.20
Elective           18914.81               4.59           1.07
```

Insurance Type Analysis

In [16]:
```
hospital_data.loc[hospital_data['Insurance_Type'] == 'Self-Pay','Insurance_Coverage'] = 0
```

```
hospital_data['Coverage_Percentage'] = (hospital_data['Insurance_Coverage']/ hospital_data['Total_Charges']) * 100

df = hospital_data.groupby('Insurance_Type').agg(
Avg_Coverage=('Insurance_Coverage', 'mean'),
Avg_Out_of_Pocket=('Patient_Out_of_Pocket', 'mean'),
Coverage_Percentage=('Coverage_Percentage', 'mean'))
df = df.sort_values(by='Avg_Coverage', ascending=False)
df[['Avg_Coverage', 'Avg_Out_of_Pocket', 'Coverage_Percentage']] = df[['Avg_Coverage', 'Avg_Out_of_Pocket', 'Coverage_Percenta
print(df)
```

```
                   Avg_Coverage  Avg_Out_of_Pocket  Coverage_Percentage
Insurance_Type
Medicaid               22455.47            9616.05                70.48
HMO                    21621.01            4159.71                84.92
PPO                    21385.26            5287.73                79.89
Medicare               19176.55            8796.00                70.40
Private Insurance      18770.21            4008.60                82.66
Self-Pay                   0.00           29109.58                 0.00
```

I grouped the data by department and severity level to see the level of readmission and complication in each department

In [17]:
```
df = hospital_data.groupby(['Department', 'Severity_Level']).agg(
    Readmission_rate=('Readmitted_30Days', 'sum'),
    complication_rate=('Complications', 'sum'))
print(df)
```

```
                                Readmission_rate  complication_rate
Department        Severity_Level
Cardiology        Critical                      0                  1
                  High                          0                  3
                  Low                           4                  2
                  Moderate                      2                  4
Emergency         Critical                      2                  2
                  High                          2                  4
                  Low                           0                  0
                  Moderate                      2                  1
General Surgery   Critical                      1                  0
                  High                          0                  0
                  Low                           3                  1
                  Moderate                      0                  1
Internal Medicine Critical                      0                  1
                  High                          0                  3
                  Low                           2                  0
                  Moderate                      1                  1
Neurology         Critical                      0                  1
                  High                          1                  1
                  Low                           2                  0
                  Moderate                      0                  0
Oncology          Critical                      2                  0
                  High                          2                  0
                  Low                           1                  1
                  Moderate                      2                  1
Orthopedics       Critical                      0                  0
                  High                          1                  1
                  Low                           1                  1
                  Moderate                      0                  0
Pediatrics        High                          1                  0
                  Low                           1                  0
                  Moderate                      0                  0
```

I decide to see how the severity level in each department affects the length of stay

In [60]:
```
df = hospital_data.groupby(['Department', 'Severity_Level']).agg(
    length_of_stay=('Length_of_Stay_Days', 'mean'))
df = df.sort_values(by='Department', ascending=True)
```

```python
df['length_of_stay'] = df['length_of_stay'].round(2)
print(df)
```

```
                                length_of_stay
Department        Severity_Level
Cardiology        Critical                15.00
                  High                    12.20
                  Low                      4.00
                  Moderate                 6.05
Emergency         Critical                13.25
                  High                     9.15
                  Low                      1.00
                  Moderate                 4.89
General Surgery   Critical                14.00
                  High                    11.00
                  Low                      4.00
                  Moderate                 6.50
Internal Medicine Low                      2.40
                  Moderate                 3.67
                  Critical                15.33
                  High                     9.14
Neurology         Critical                11.33
                  High                     8.75
                  Low                      2.25
                  Moderate                 4.92
Oncology          Moderate                 5.07
                  Low                      2.44
                  Critical                13.67
                  High                     7.43
Orthopedics       Critical                16.00
                  High                     9.75
                  Low                      4.11
                  Moderate                 7.00
Pediatrics        Low                      2.20
                  High                     8.83
                  Moderate                 4.00
```

I had to group the wait time into categories so i can see if the wait time and staff ratio affects the mean satisfaction rate

In [19]:
```python
wait_bins = [0, 2, 4, 12]
wait_labels = ['≤2hrs', '2-4hrs', '>4hrs']
```

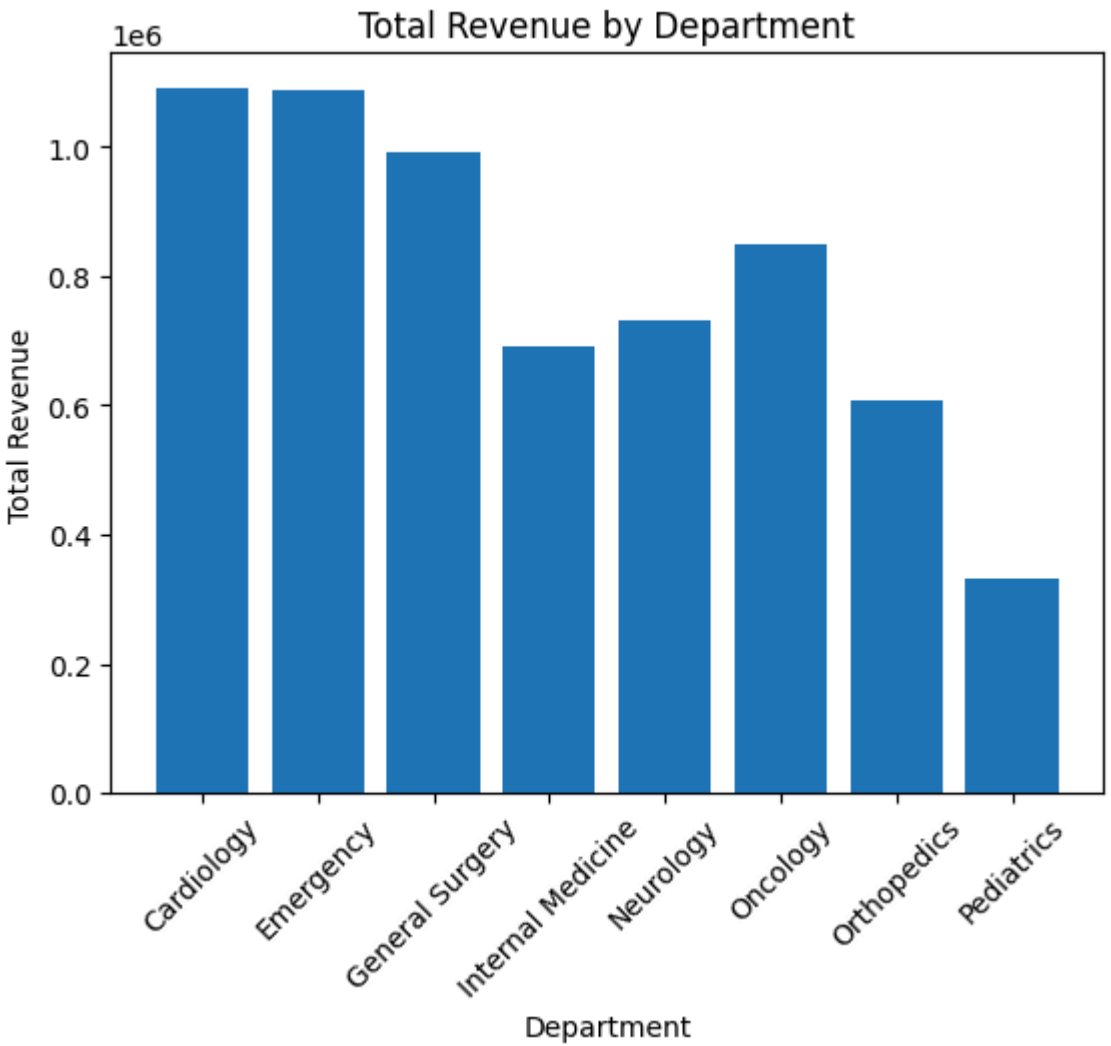
```python
hospital_data['Wait_Time_Category'] = pd.cut(hospital_data['Wait_Time_Hours'],bins=wait_bins,
labels=wait_labels,right=False)

df = hospital_data.groupby(['Department', 'Wait_Time_Category']).agg(
Avg_Satisfaction=('Patient_Satisfaction_Score', 'mean'),
Avg_Staff_Ratio=('Staff_to_Patient_Ratio', 'mean'))

df[['Avg_Satisfaction', 'Avg_Staff_Ratio']] = df[['Avg_Satisfaction', 'Avg_Staff_Ratio']].round(2)


print(df)
```

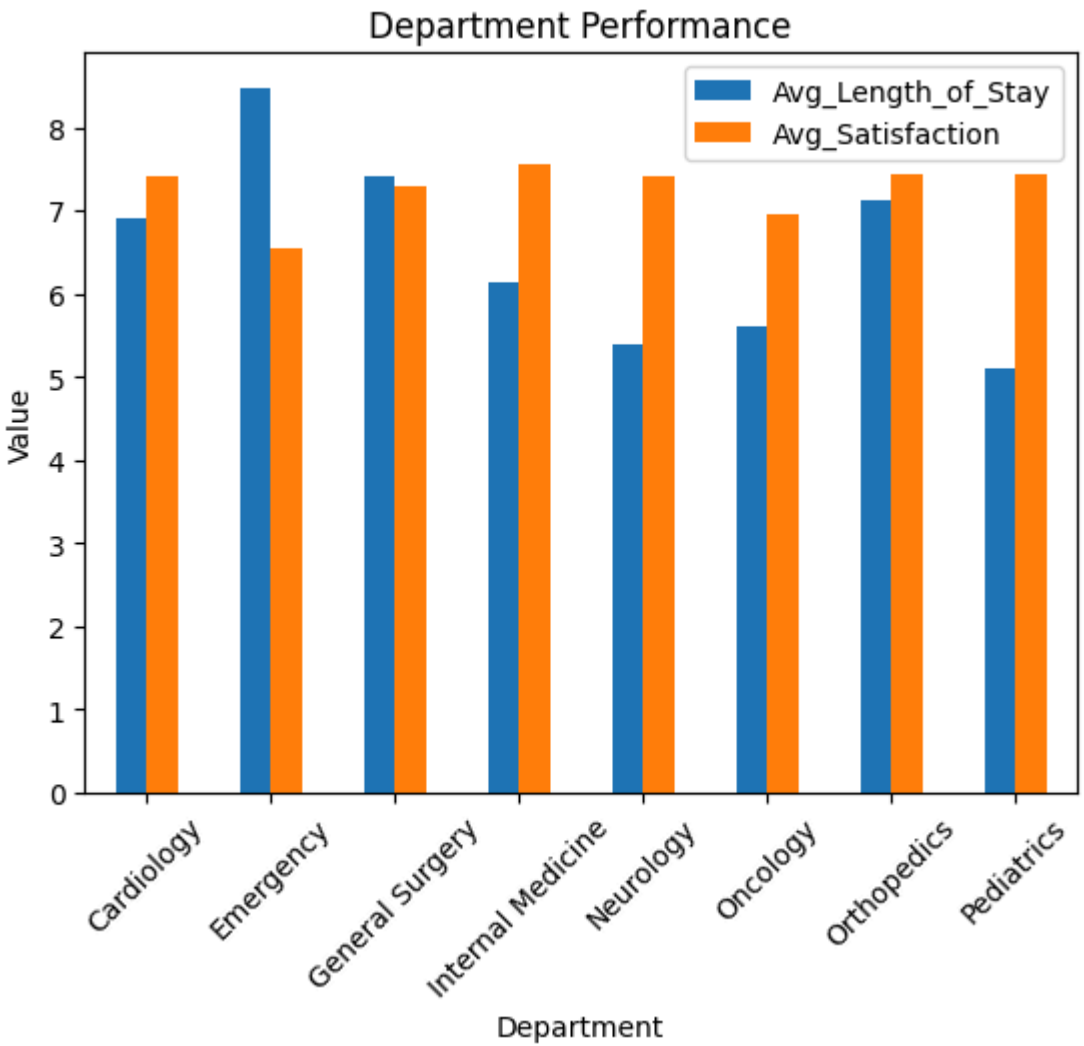
```
                                   Avg_Satisfaction  Avg_Staff_Ratio
Department        Wait_Time_Category
Cardiology        ≤2hrs                        7.31             2.07
                  2-4hrs                       7.45             2.48
                  >4hrs                        8.45             3.00
Emergency         ≤2hrs                        5.68             2.73
                  2-4hrs                       6.63             2.39
                  >4hrs                        6.80             3.07
General Surgery   ≤2hrs                        7.55             2.12
                  2-4hrs                       7.42             2.84
                  >4hrs                        5.95             3.30
Internal Medicine ≤2hrs                        7.31             2.31
                  2-4hrs                       8.39             2.37
                  >4hrs                        7.14             3.22
Neurology         ≤2hrs                        7.89             2.15
                  2-4hrs                       7.49             2.34
                  >4hrs                        6.36             3.33
Oncology          ≤2hrs                        6.72             2.21
                  2-4hrs                       7.34             2.61
                  >4hrs                        7.03             2.64
Orthopedics       ≤2hrs                        7.51             2.13
                  2-4hrs                       7.60             1.80
                  >4hrs                        7.08             2.90
Pediatrics        ≤2hrs                        6.75             1.90
                  2-4hrs                       7.69             2.30
                  >4hrs                        8.30             2.53
```

I made befitting chart for all the grouped data to make it easier to analyse

In [20]:
```
import matplotlib as mpl
import matplotlib.pyplot as plt
```

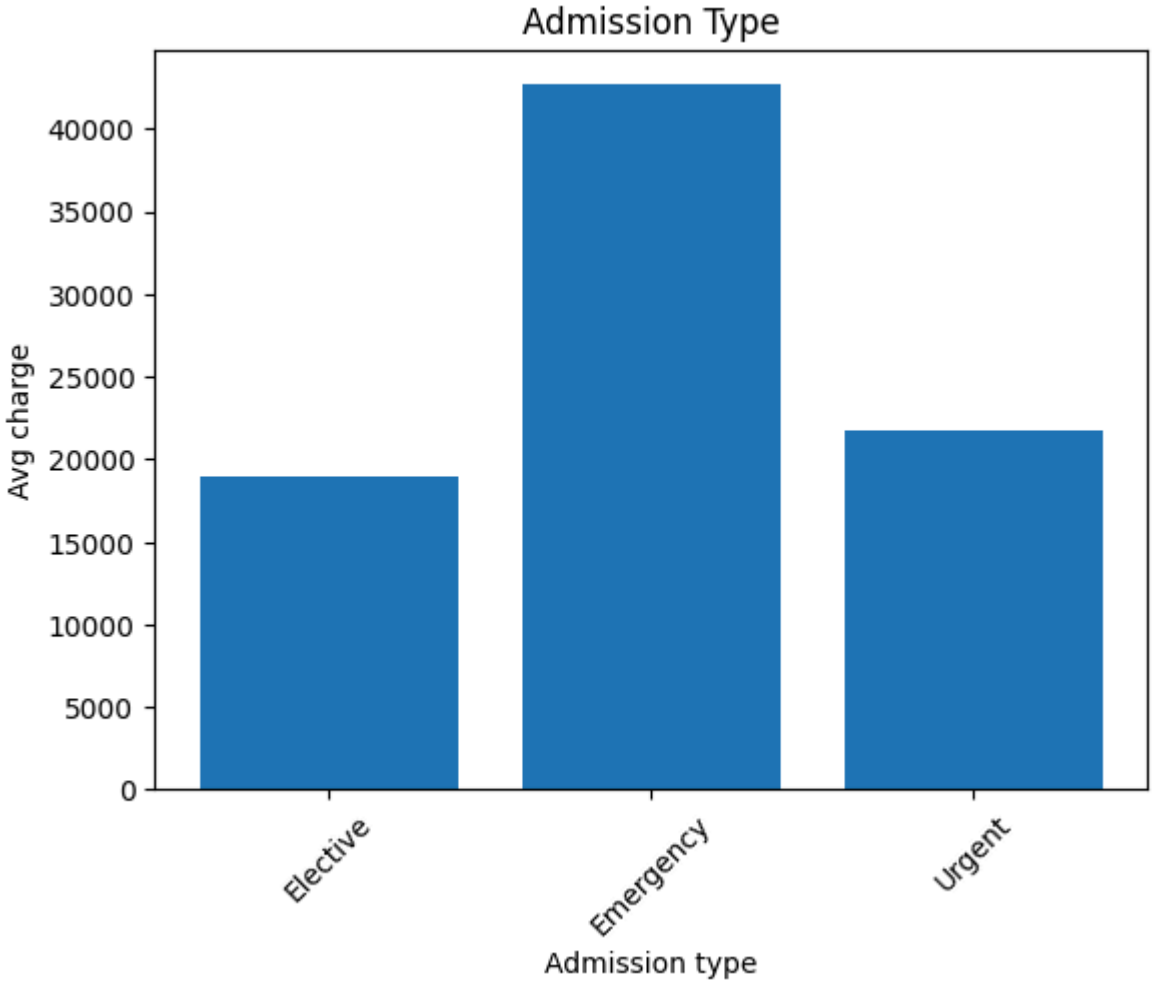
In [68]:
```
df = hospital_data.groupby('Department').agg(
    Patient_Count=('Department', 'size'),
    Avg_Length_of_Stay=('Length_of_Stay_Days', 'mean'),
    Total_Revenue=('Total_Charges', 'sum'),
    Avg_Satisfaction=('Patient_Satisfaction_Score', 'mean'))

plt.bar(df.index, df['Total_Revenue'])

plt.xlabel('Department')
plt.ylabel('Total Revenue')
plt.title('Total Revenue by Department')

plt.xticks(rotation=45)
plt.show()

df[['Avg_Length_of_Stay', 'Avg_Satisfaction']].plot(kind='bar')
plt.xlabel('Department')
plt.ylabel('Value')
plt.title('Department Performance')

plt.xticks(rotation=45)
plt.show()
```

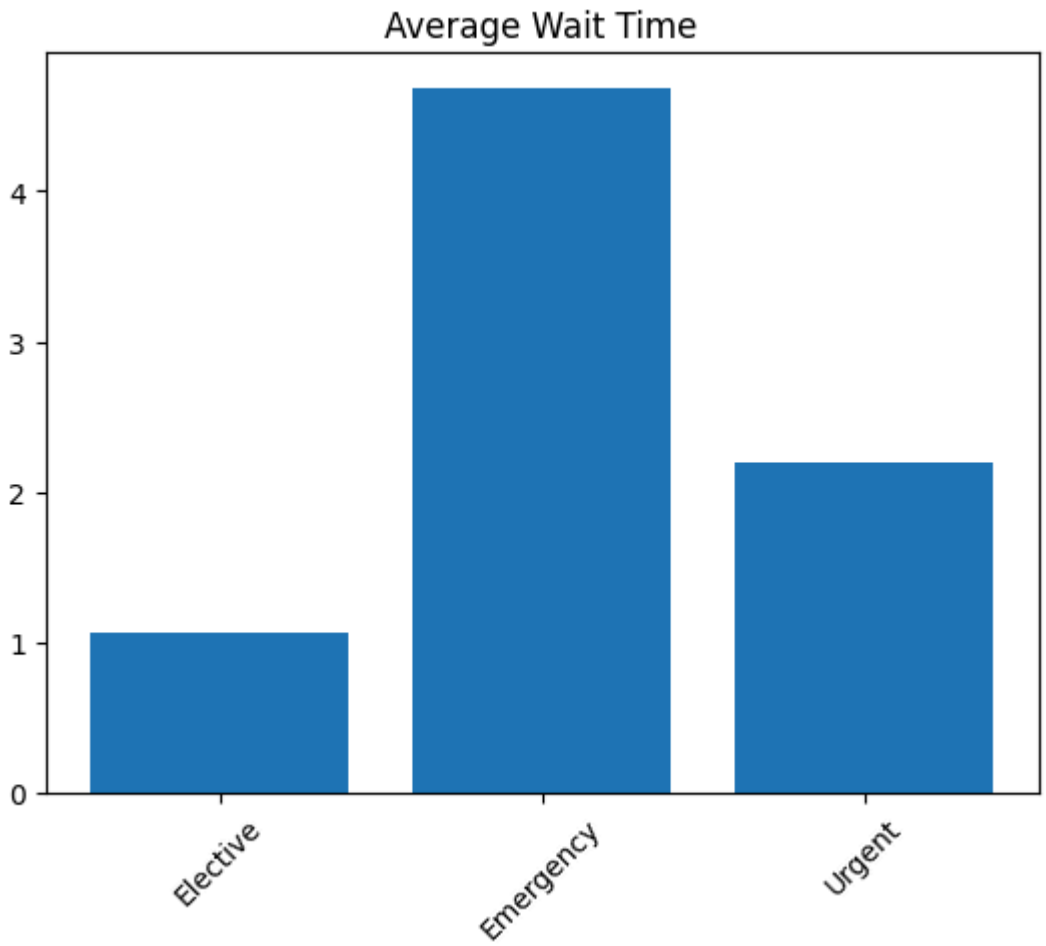

Total Revenue by Department
1e6
1.0
0.8
0.6
0.4
0.2
0.0
Total Revenue
Cardiology
Emergency
General Surgery
Internal Medicine
Neurology
Oncology
Orthopedics
Pediatrics
Department



In [44]:
```
df = hospital_data.groupby('Admission_Type').agg(
Avg_Charge=('Total_Charges', 'mean'),
Avg_Length_of_Stay=('Length_of_Stay_Days', 'mean'),
Avg_wait_time=('Wait_Time_Hours', 'mean'))
plt.bar(df.index, df['Avg_Charge'])

plt.xlabel('Admission type')
plt.ylabel('Avg charge')
```

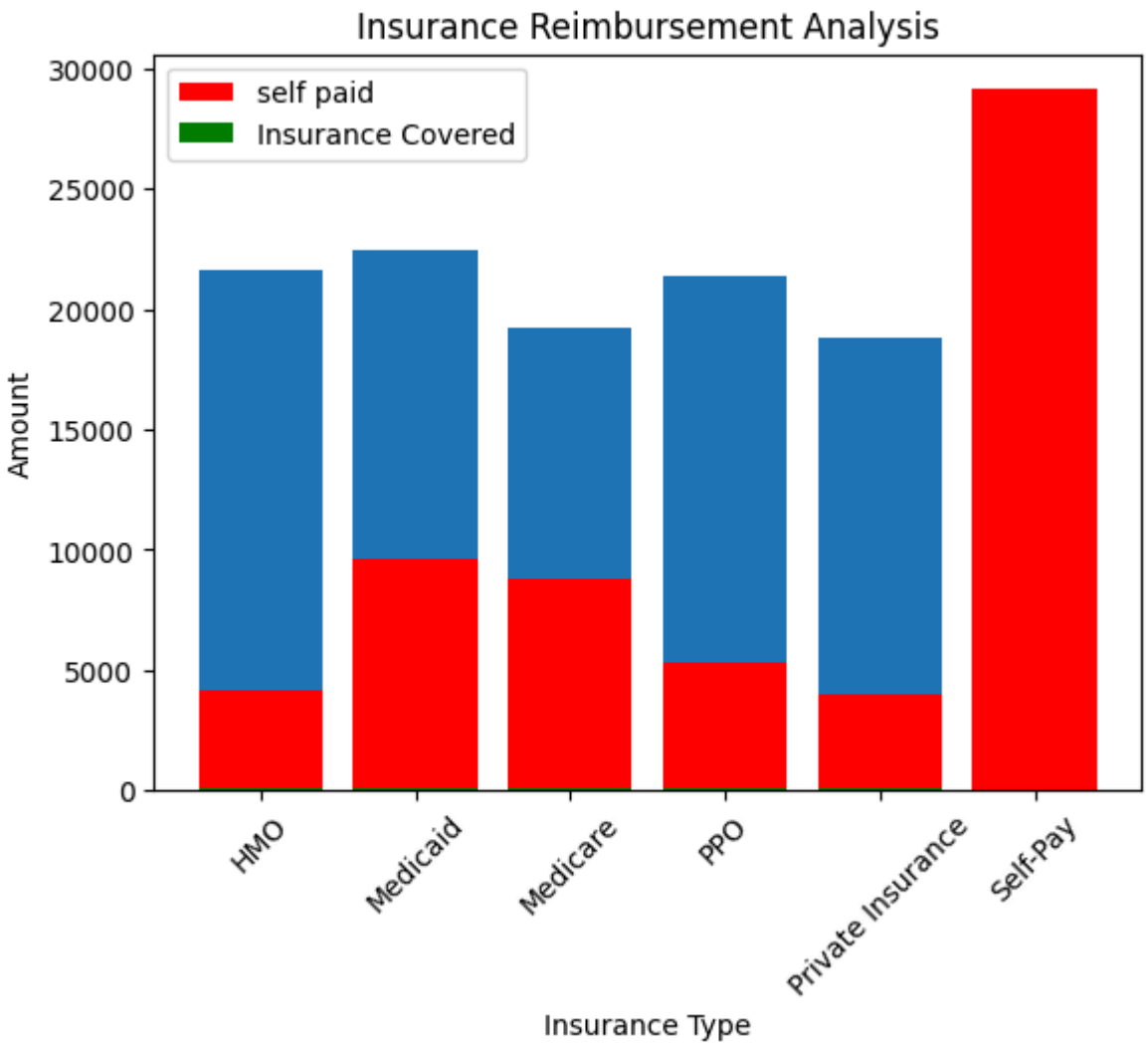
```
plt.title('Admission Type')

plt.xticks(rotation=45)
plt.show()

plt.bar(df.index, df['Avg_wait_time'])
plt.title('Average Wait Time')
plt.xticks(rotation=45)
plt.show()
```



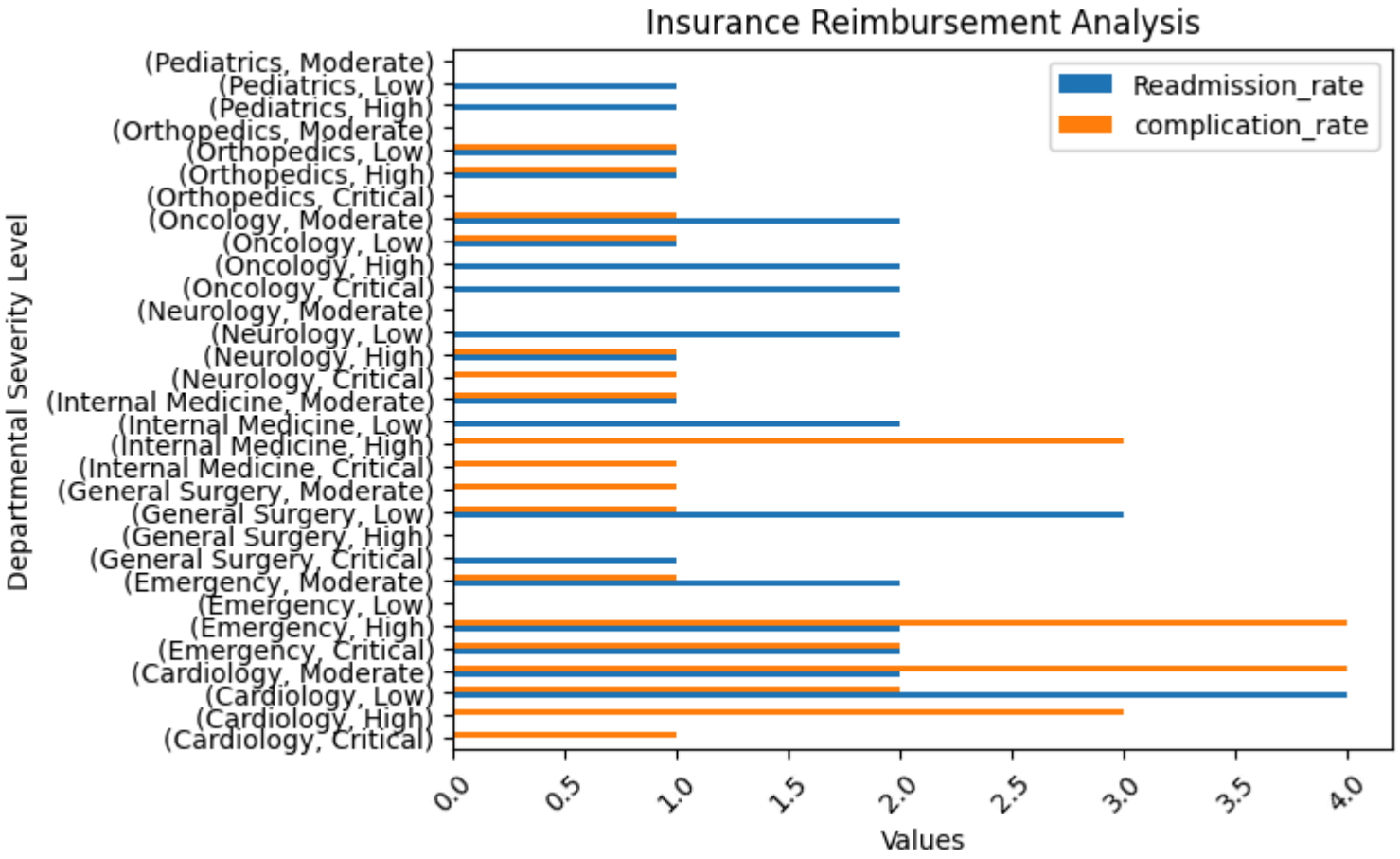


In [66]:
```
hospital_data.loc[hospital_data['Insurance_Type'] == 'Self-Pay','Insurance_Coverage'] = 0

hospital_data['Coverage_Percentage'] = (hospital_data['Insurance_Coverage']/ hospital_data['Total_Charges']) * 100

df = hospital_data.groupby('Insurance_Type').agg(
Avg_Coverage=('Insurance_Coverage', 'mean'),
Avg_Out_of_Pocket=('Patient_Out_of_Pocket', 'mean'),
Coverage_Percentage=('Coverage_Percentage', 'mean'))

plt.bar(df.index, df['Avg_Coverage'])
plt.bar(df.index, df['Avg_Out_of_Pocket'], label='self paid', color='r')
```

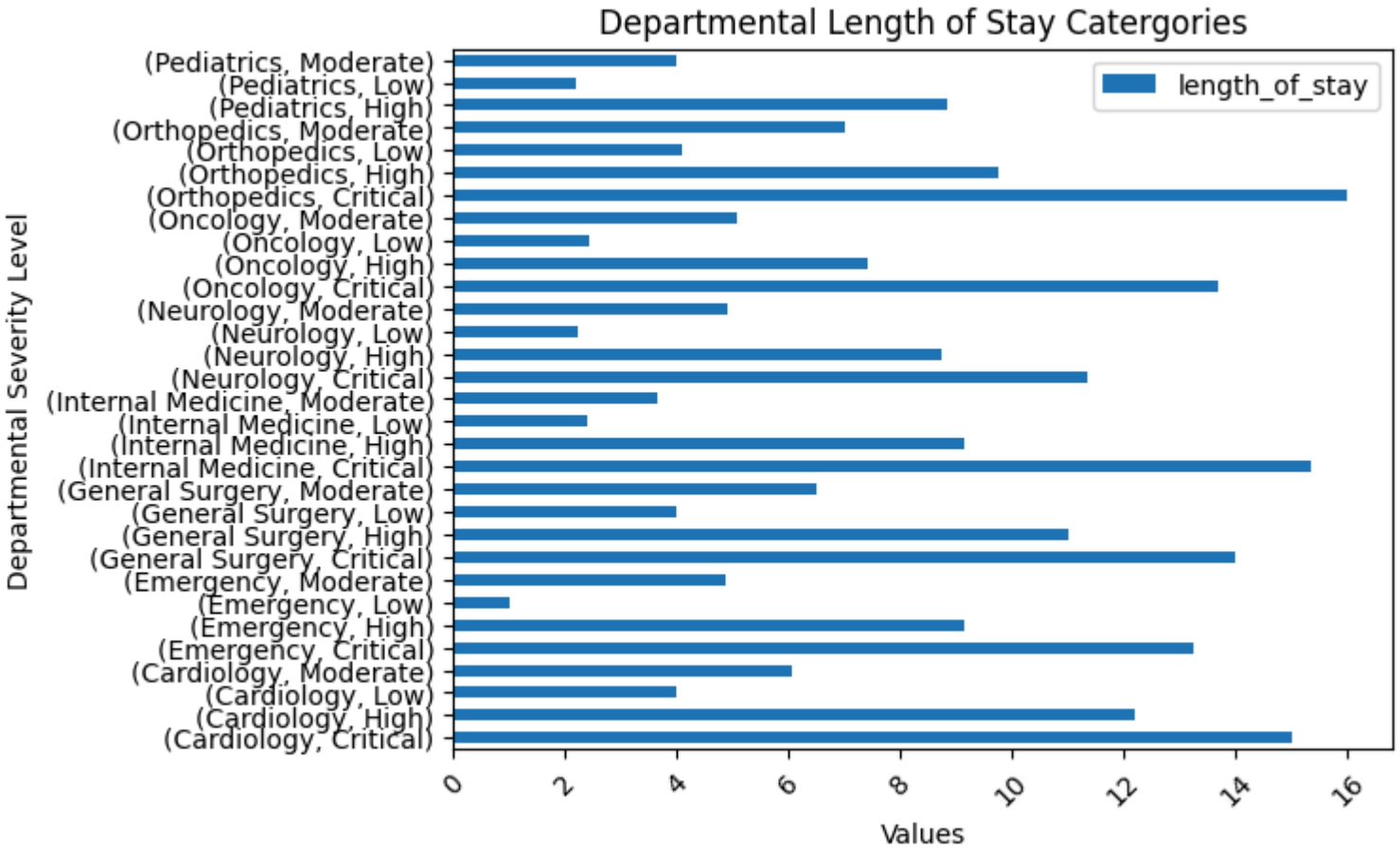
```
plt.bar(df.index, df['Coverage_Percentage'], label='Insurance Covered',color='g')

plt.xlabel('Insurance Type')
plt.ylabel('Amount')
plt.title('Insurance Reimbursement Analysis')
plt.legend()
plt.xticks(rotation=45)
plt.show()
```



In [57]:
```
df = hospital_data.groupby(['Department', 'Severity_Level']).agg(
    Readmission_rate=('Readmitted_30Days', 'sum'),
    complication_rate=('Complications', 'sum'))

df[['Readmission_rate', 'complication_rate']].plot(kind='barh')
plt.xlabel('Values')
plt.ylabel('Departmental Severity Level')
plt.title('Insurance Reimbursement Analysis')
plt.legend()
plt.xticks(rotation=45)
plt.show()
```

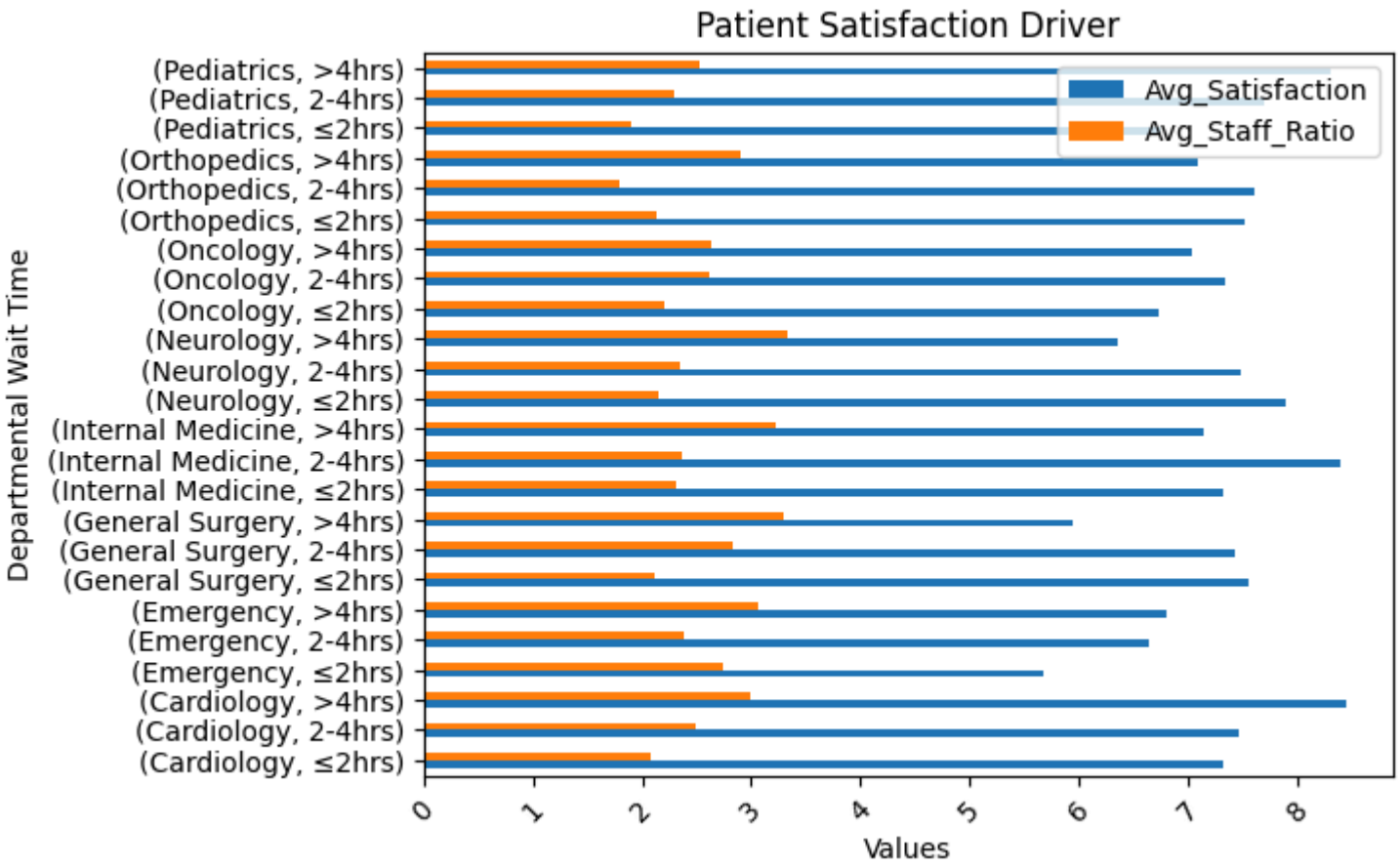


In [63]:
```
df = hospital_data.groupby(['Department', 'Severity_Level']).agg(
    length_of_stay=('Length_of_Stay_Days', 'mean'))
df['length_of_stay'].plot(kind='barh')
plt.xlabel('Values')
plt.ylabel('Departmental Severity Level')
plt.title('Departmental Length of Stay Catergories')
plt.legend()
plt.xticks(rotation=45)
plt.show()
```



In [65]:
```
wait_bins = [0, 2, 4, 12]
wait_labels = ['≤2hrs', '2-4hrs', '>4hrs']
```

```
hospital_data['Wait_Time_Category'] = pd.cut(hospital_data['Wait_Time_Hours'],bins=wait_bins,
labels=wait_labels,right=False)

df = hospital_data.groupby(['Department', 'Wait_Time_Category']).agg(
Avg_Satisfaction=('Patient_Satisfaction_Score', 'mean'),
Avg_Staff_Ratio=('Staff_to_Patient_Ratio', 'mean'))

df[['Avg_Satisfaction', 'Avg_Staff_Ratio']].plot(kind='barh')
plt.xlabel('Values')
plt.ylabel('Departmental Wait Time')
plt.title('Patient Satisfaction Driver')
plt.legend()
plt.xticks(rotation=45)
plt.show()
```



The analysis showed that Length of Stay is the primary driver of the hospital total charges, directly impacting the financial KPI, while wait time and staffing levels does not necessarily influence patient satisfaction rating. Cardiology department has the highest patient count and contributes the most to the total charges, while pediatrics has the least patient count and also the least total charges contribution.